

Lebovitz, Jacob

Data c104

2 May 2024

Analysis Essay #2

Within a datafied world, data has become connected to many aspects of human life, including labor, research, and economics. While data can be a powerful tool, it can also expose injustice and enhance unhealthy social dynamics. Furthermore, data can be crafted to narrate a story of success, righteousness, and growth, leaving out stories of oppression and muting the voices of the powerless. We are in a world of constant technological growth, supported by Silicon Valley culture which beautifies technology as a solution for personal and societal issues. Often, projects, like the one described with Arkenstone, ignore the ethical stakes of their product and research. This project aims to increase the productivity and overall success of this startup by analyzing the labor force and learning about worker productivity. However, the lack of attention to the social implications of the project have called for an examination of the ethical stakes of such data-centered research, specifically the use of demographics in machine algorithms, missing institutional regulation on subjects, and increased exposure creating means for exploitation.

In this project, it is mentioned that worker demographic information is available to be used in conjunction with efficiency data in order to determine optimal placement. However, it is evident that machines can correlate and inaccurately link demographic information with performance ability due to pattern matching and historic social systems that may have created segregation in the workplace. The COMPAS algorithm, a system that used racial proxies to determine the risk of recidivism among defendants, was shown to misclassify more black people as likely to recommit compared to white people. These algorithms were put in place as a

seemingly more objective way to organize the incarceration system, yet its bias shows that the machine learning algorithm used data drawn from a social context created by discriminatory people. The results of COMPAS have initiated a cycle of performativity by assigning more black people a higher score, leading to more surveillance and higher chance of them being observed committing a crime. The same feedback mechanism is a risk in this research project; if information like race, education level, and gender are being used in a model of performance analysis, it is entirely possible for the model to associate personal data with their work ability. This could create a performative cycle of marginalization by showing Arkenstone's managers how certain demographics work compared to others, which is not an accurate optimization model because this model could be based on data collected when there were unjust social barriers. In the article "The Collision of Demographics, Automation and Inequality," it is mentioned that an aging population is reducing the number of viable workers, creating a narrative pushing automation which ignores many other aspects of the issue, including ideas of reducing personal consumption or promoting employment longevity. This article parallels the narratives of opportunity being told in this research group, but ignores the ethical inequality stakes and performativity that these narratives can reinforce.

Additionally, it is mentioned that the PI did not report this research to the IRB, a risk that could result in unethical treatment of subjects and informational harm. While institutional regulation has been critiqued because of its broad language, the IRB can be successful in holding research to a base standard that protects human subjects. The choice to not work with this organization is legal because many data-focused projects are able to justify a disconnect between the subject because their data was collected prior and with consent. However, using personal data without interaction with that person can remove their ability to self-represent, forcing these

individuals to be spoken for only through their data. Additionally, the exposure of information can lead to informational harm, as the use and analysis of such data, which is connected to a national identification system, makes certain marginalized demographics more vulnerable. Furthermore, it is important to consider how the benefits of this research will be distributed, as it could create employment opportunities and challenges for different groups, similar to the Tuskegee Syphilis study, which did create scientific progress, but significantly harmed the black human subjects, violating ethical codes. An article published in Sage Journals mentions that “datasets and algorithms have historical, material specificity that is laden with political and ethical values” (Metcalf, Crawford 2016). In the project with Arkenstone, it is evident that the big data are interconnected to the political and social atmosphere in this South American country, making “these values become both more evident and more consequential” (Metcalf, Crawford 2016).

Finally, the use of data to promote profits is a result of accumulated capitalist spirit. With the CEO and PI graduating from the Bay Area, it is probable that they have experienced the Silicon Valley mindset of techno-solutionism and data capitalism. It is also important to see that “the shift towards data capital takes advantage of the ideological and regulatory groundwork that has been laid since at least the 1980s to create a political economic landscape conducive to finance capitalism,” (Sadowski 2019) illuminating that companies using data as a means of capital expansion should assess the same ethical issues that capitalism exploits. Using personal data in research as a way to optimize performance raises the question of who is benefiting from the research, a similar dilemma to the diabetes study with the Gila River community. This system of research allows for the PI to control the narrative told, which is one of liberation and success. However, datafied labor is a socio-technical system co-produced with technological innovation,

so it is important to study how technology can unequally disrupt the workforce, and why that narrative is silenced through capitalism.

It is essential to discuss the ethical stakes of this project which uses personal data and impacts the labor opportunities of displaced workers. In a society reliant on personal data in research, economics, and business, the definition of a human subject project is due to expand, enhancing the attention to the ethical stakes of such endeavors.

Bibliography

- Harris, Karen, Austin Kimson, and Andrew Schwedel. "Bain Report Labor 2030: The Collision of Demographics, Automation and Inequality Executive Summary," 2018.
https://www.explodingafrica.com/wp-content/uploads/2018/04/Labor-2030.-The-Collision-of-Demographics-Automation-and-Inequality_Bain-Company-1.pdf.
- Metcalf, Jacob, and Kate Crawford. "Where Are Human Subjects in Big Data Research? The Emerging Ethics Divide." *Big Data & Society* 3, no. 1 (January 5, 2016): 205395171665021. <https://doi.org/10.1177/2053951716650211>.
- Sadowski, Jathan. "When Data Is Capital: Datafication, Accumulation, and Extraction." *Big Data & Society* 6, no. 1 (January 7, 2019): 1–12.
<https://doi.org/10.1177/2053951718820549>.